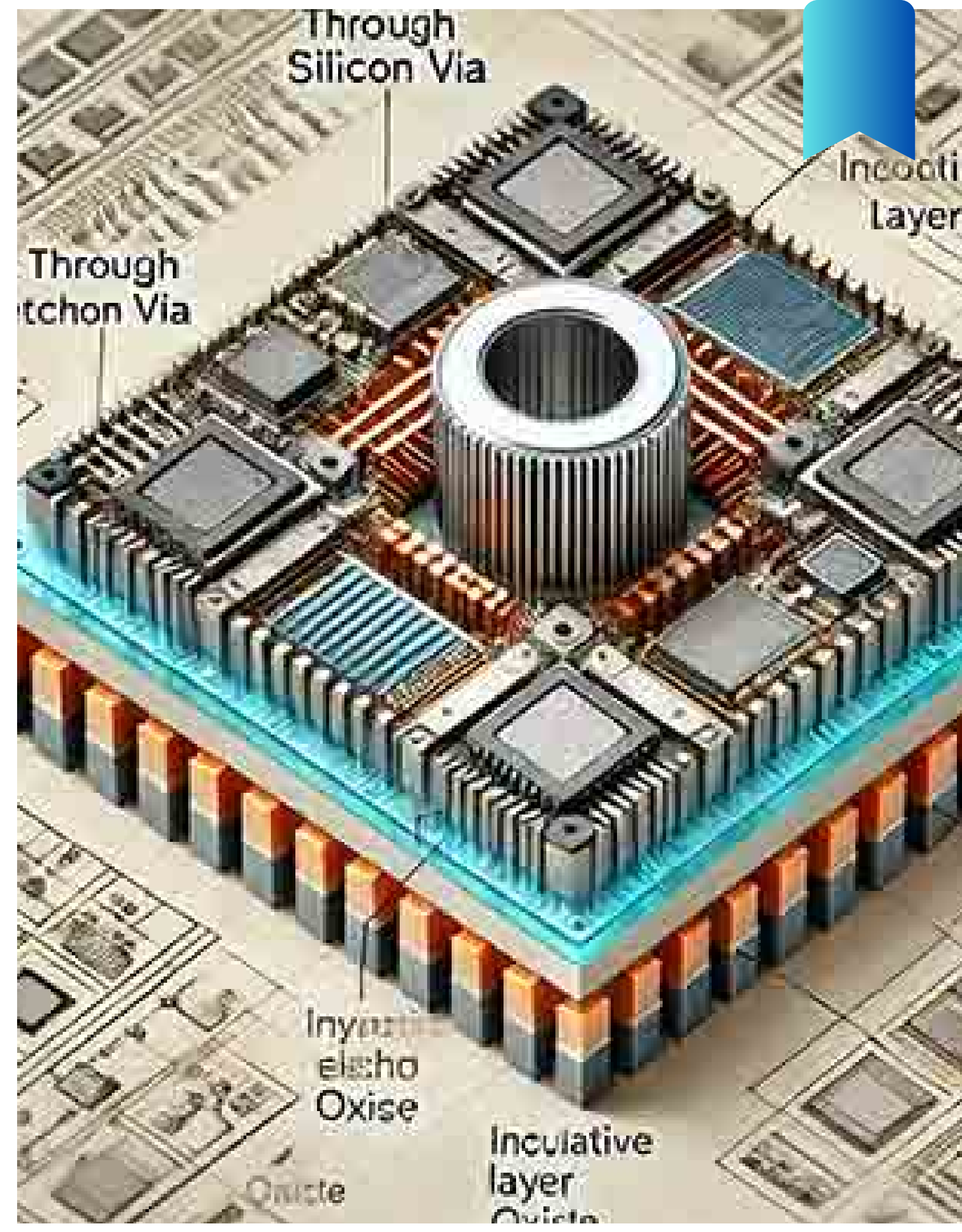


Design Equations:

RLGC Equations of Defect-free and Defected TSV

By: Kamineni Dhana Sri Keerthi



Why Choose Cylindrical TSV?

➤ Easier Fabrication:

- TSVs are formed using DRIE (Deep Reactive Ion Etching).
- DRIE naturally produces nearly circular holes.

➤ Uniform Electrical Field Distribution:

- Electric field lines spread symmetrically around the TSV.
- Reduced field concentration at edges.

➤ Less Concentration:

- Since a cylinder has no sharp corners, the field lines doesn't get concentration.
- So it reduces Electrical Stress.

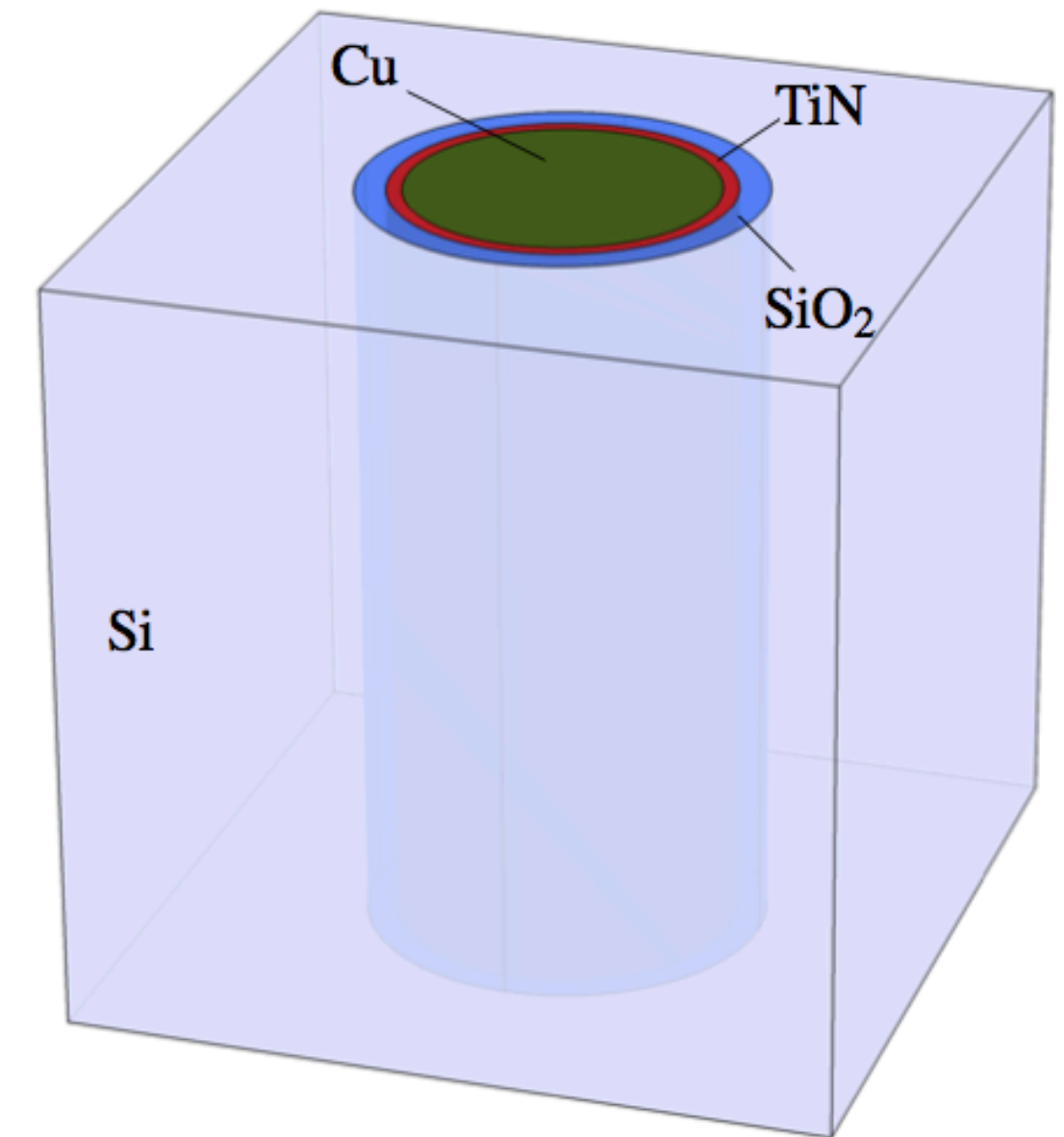


Fig.1 Cylindrical Through Silicon via

Types of Networks

Pi Network:

- Consists of one series element and two shunt element.
- The Shunt Element gets divided into half.

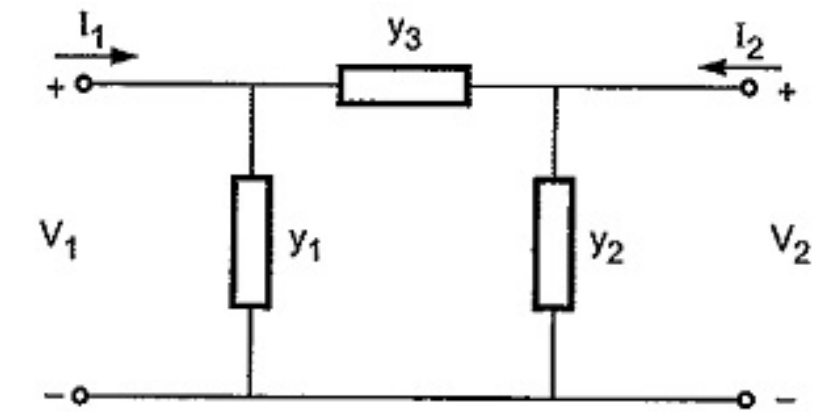


Fig.2.a. PI Network

T Network:

- Consists of two series element and one shunt element.
- The Series Element gets divided into half.

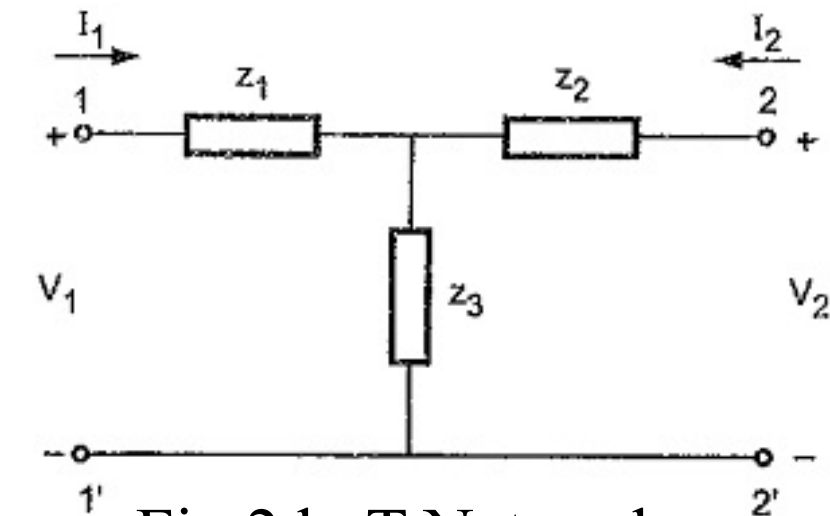


Fig.2.b. T Network

L Network:

- Consists of one series element and one shunt element.
- Shape resembles the letter L.

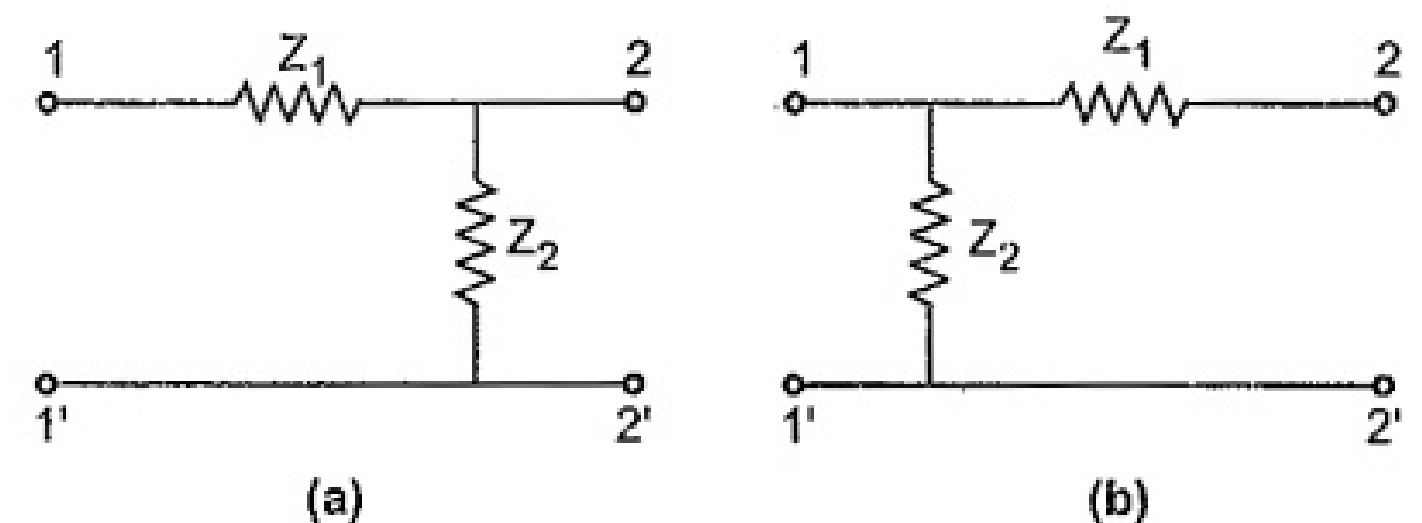


Fig.2.c. T Network

Types of Electrical Models

Lumped Model:

- Electrical components are concentrated at discrete points.
- Voltage and current are considered uniform throughout each component
- If the circuit size is much smaller than the wavelength
- Example: R,L,C.

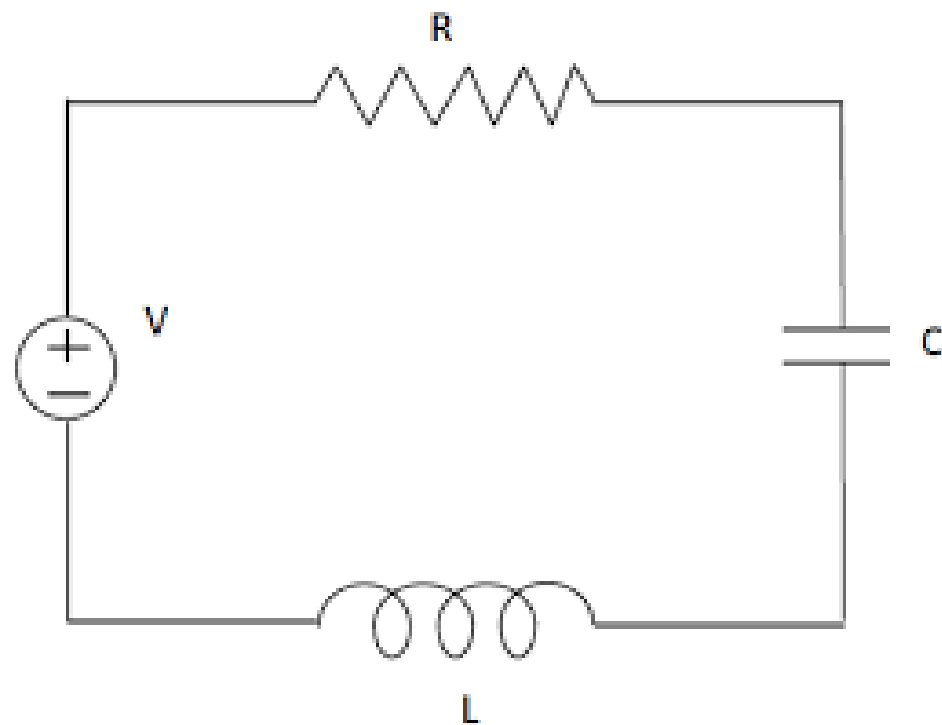


Fig.3.a. Lumped Model

Distributed Model:

- Electrical parameters are distributed continuously along the structure.
- Voltage and current vary with position and time.
- Required when physical dimensions are comparable to the signal wavelength.
- Represented using distributed resistance, inductance, capacitance, and conductance per unit length
- Example: Transmission line

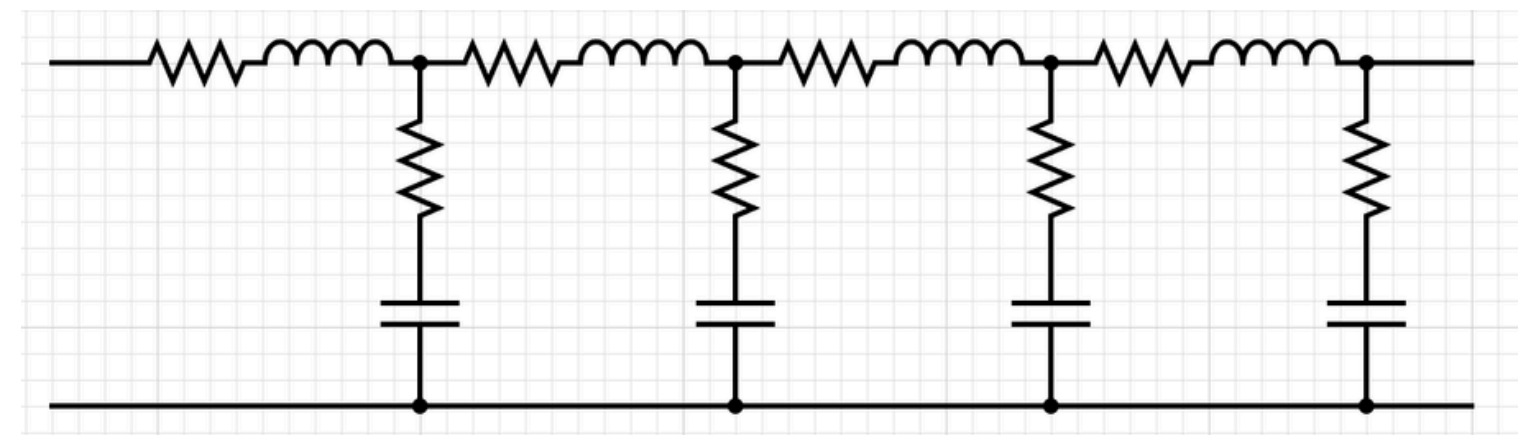


Fig.3.b. Distributed Model



Defect Free TSV Electrical Modelling

RLGC Equations



Circuit Diagram

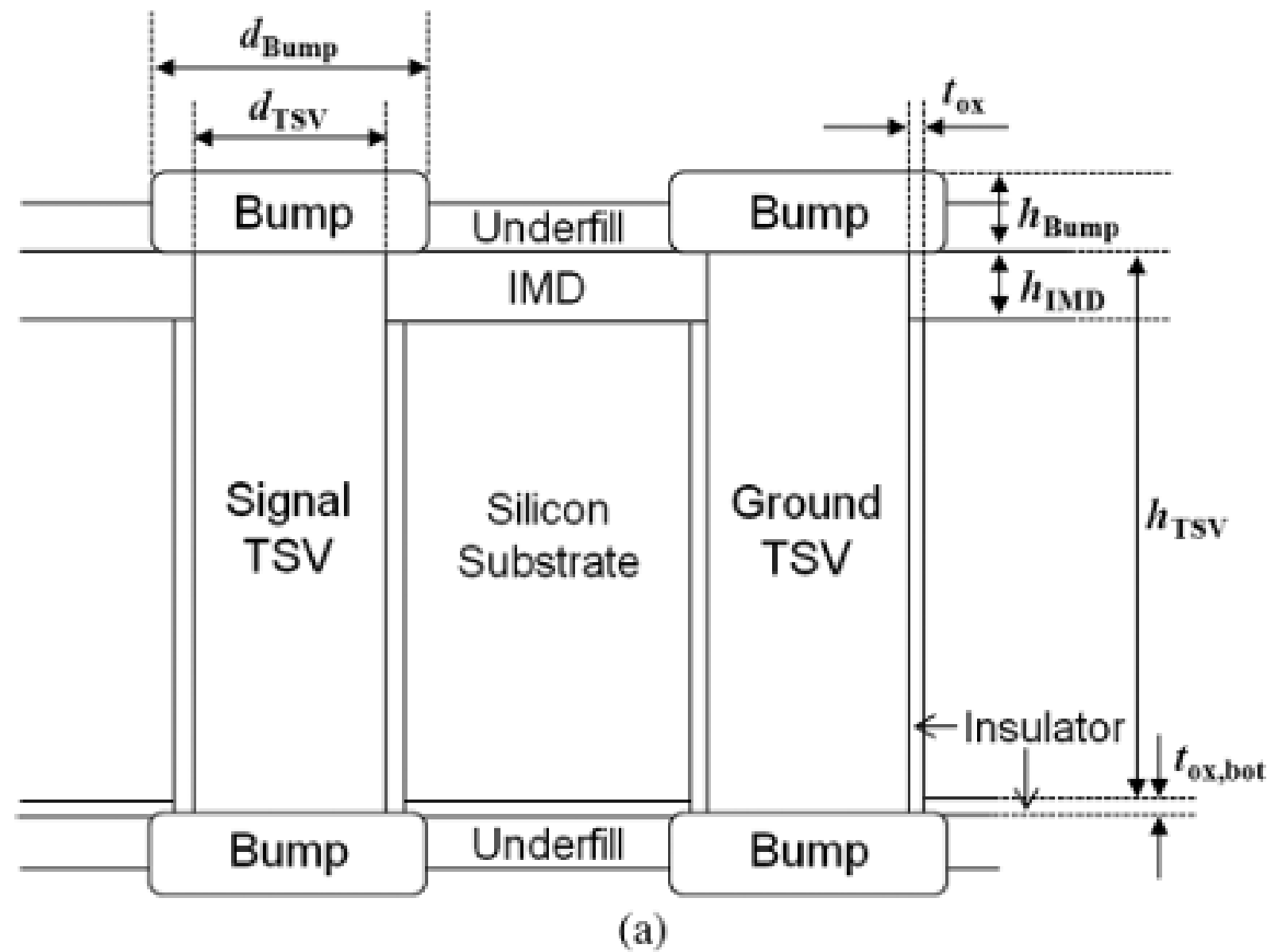


Fig.4 (a) Structure of a signal TSV and a ground TSV with bumps with the via-last process and their structural parameters⁽¹⁾

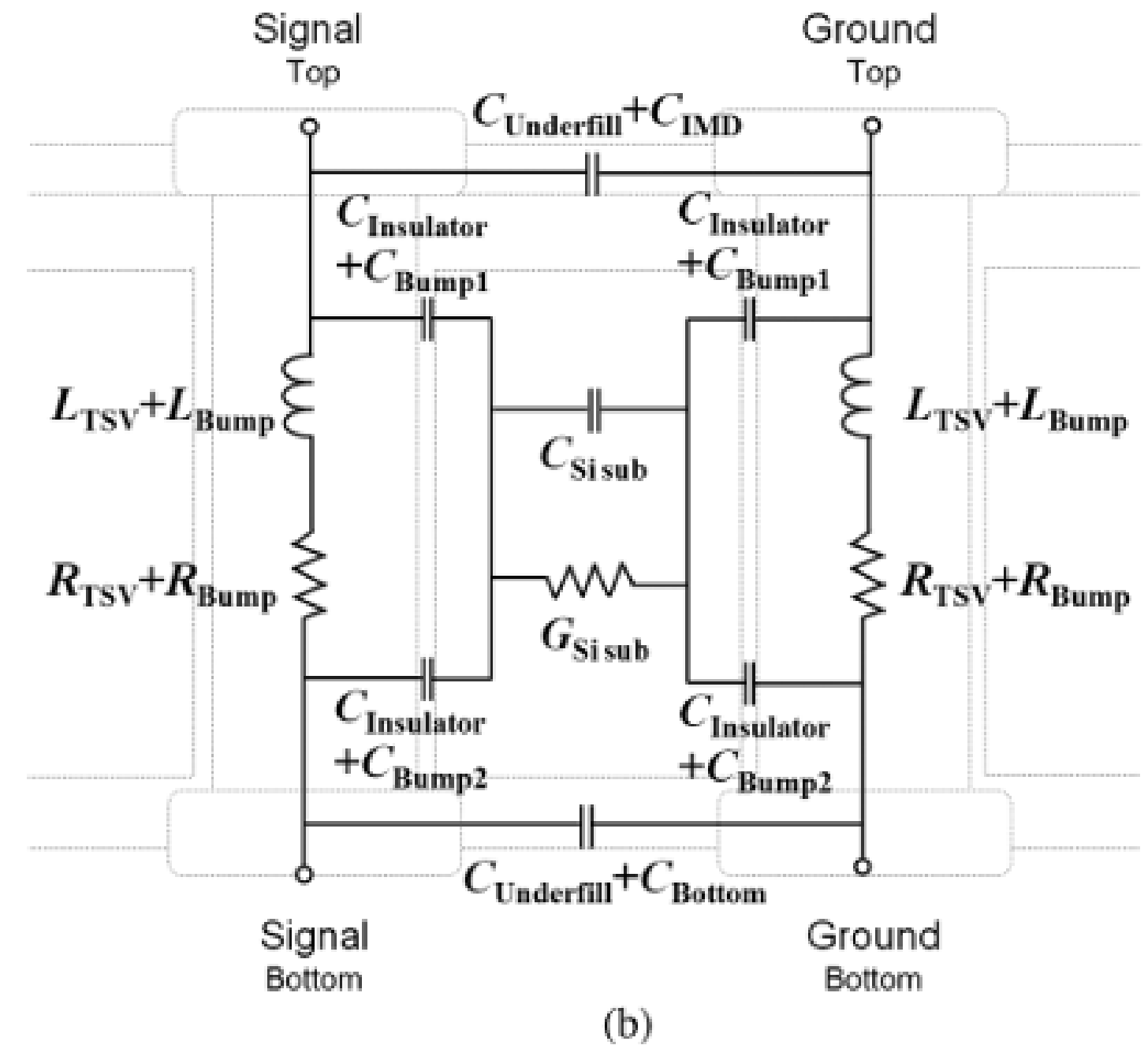


Fig.4 (b) The scalable electrical model of Fig.4(a) with labelled components⁽¹⁾



Types of Capacitances



Insulator Capacitance



Bump01 Capacitance



Bump02 Capacitance



Underfill Capacitance



IMD Capacitance



Bottom Capacitance



Substrate Capacitance



Insulator Capacitance

$$C_{\text{Insulator}} = \frac{1}{2} \left\{ 2\pi \times \varepsilon_0 \varepsilon_{r,\text{Insulator}} \times \frac{h_{\text{TSV}} - h_{\text{IMD}}}{\ln \left(\frac{d_{\text{TSV}}/2 + t_{\text{ox}}}{d_{\text{TSV}}/2} \right)} \right\} [F]. \quad (1)$$

➡ Insulator Capacitance is a Cylindrical Capacitance

➡ TSV - INSULATOR - Silicon Substrate

➡ The General Formula for a Cylindrical Capacitance is

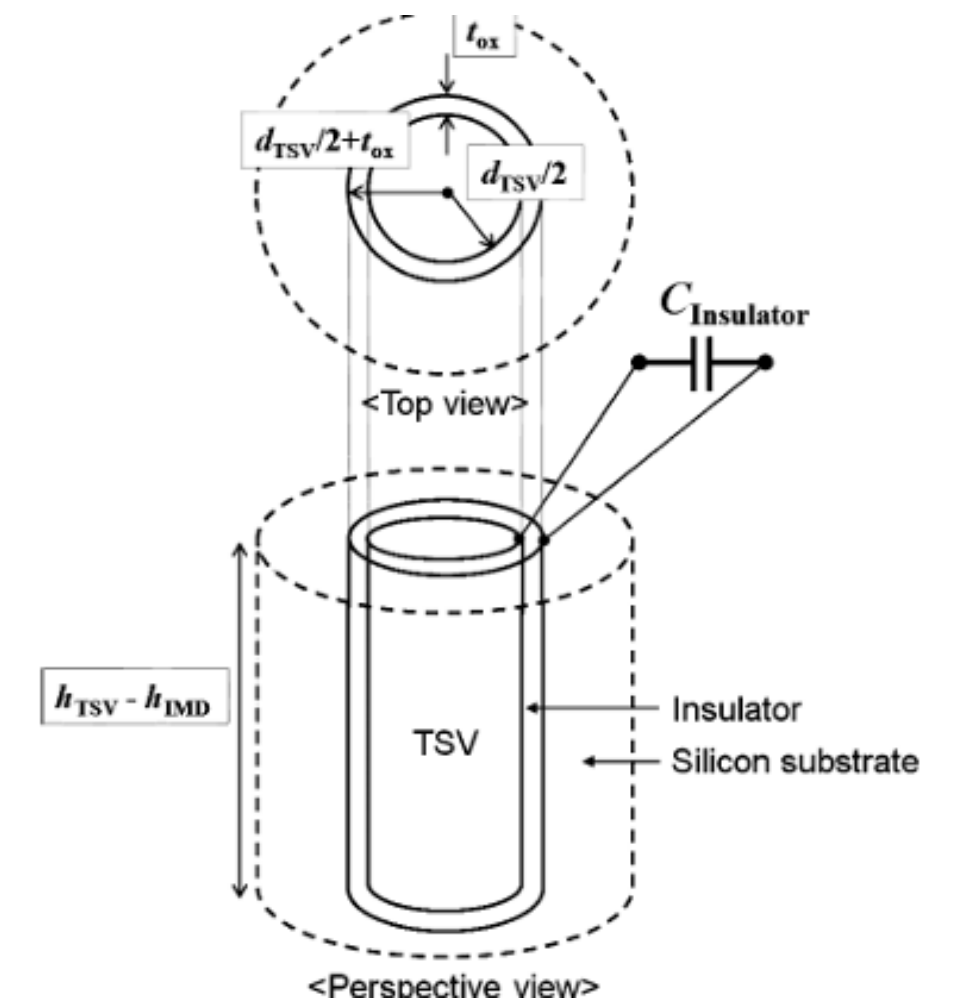


Fig.5.a. Top view and a perspective view of a TSV⁽¹⁾

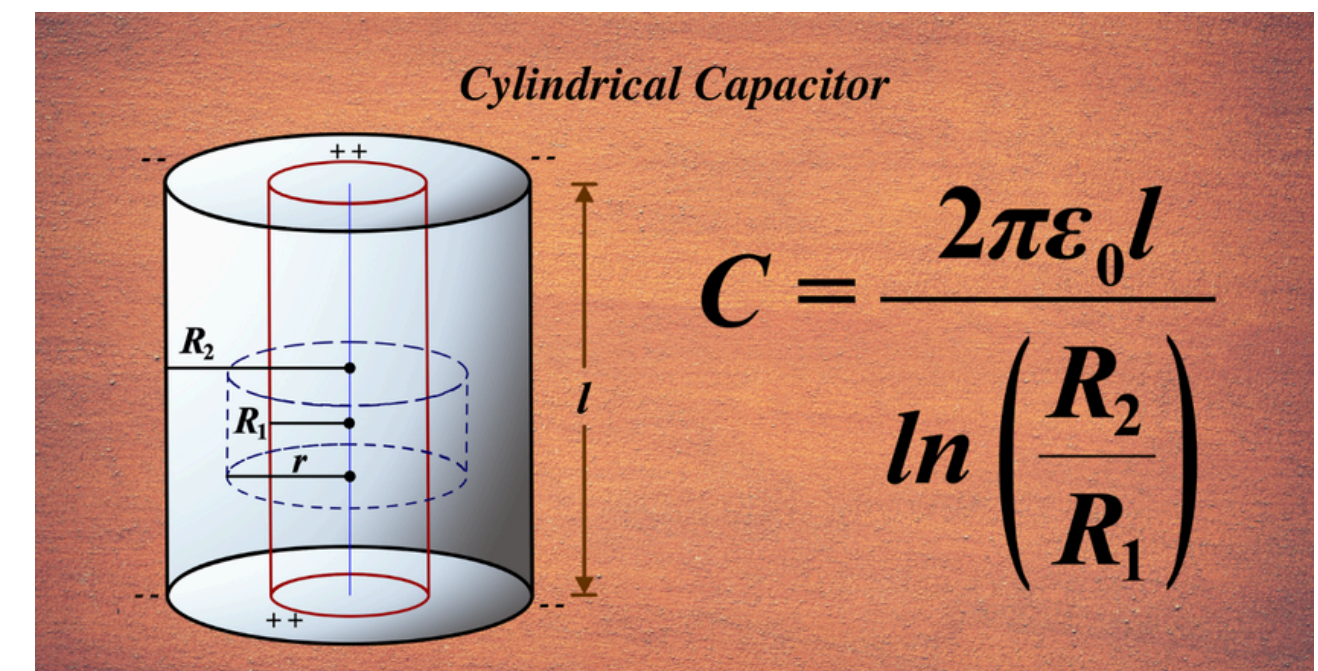


Fig.5.b. Cylindrical capacitor

Bump01 Capacitance

$$C_{\text{Bump1}} = \varepsilon_0 \varepsilon_{r,\text{IMD}} \times \frac{\pi \times \left\{ (d_{\text{Bump}}/2)^2 - (d_{\text{TSV}}/2 + t_{\text{ox}})^2 \right\}}{h_{\text{IMD}}} [F]$$

➡ Bump01 Capacitance is a Parallel Capacitance

➡ Bump - IMD - Silicon Substrate

➡ The General Formula for a Cylindrical Capacitance is

$$C = \frac{\varepsilon_0 A}{d}$$

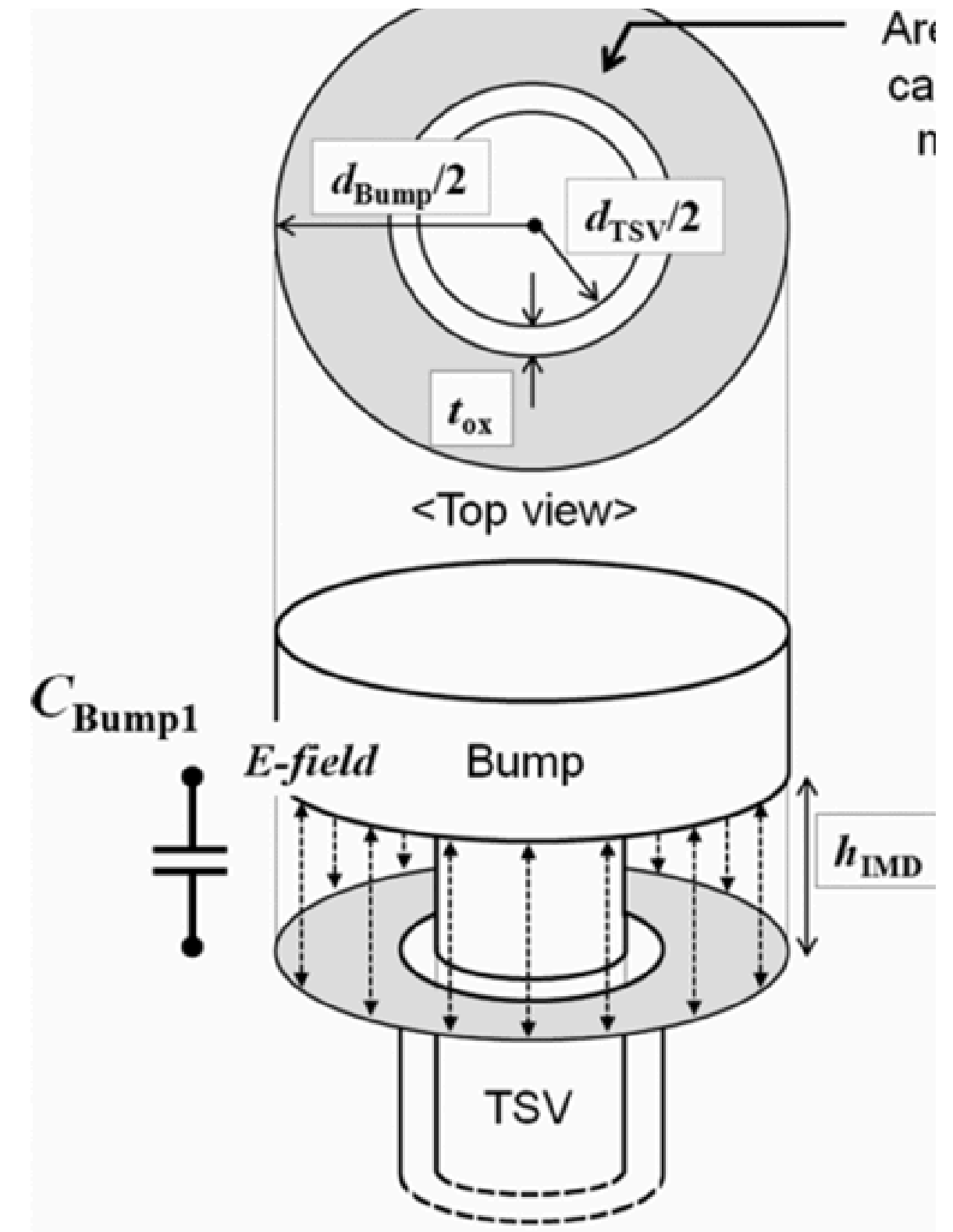


Fig.6.a. A top view and a perspective view of the upper part of the TSV ⁽¹⁾

Bump02 Capacitance

$$C_{\text{Bump2}} = \varepsilon_0 \varepsilon_{r,\text{ox}} \times \frac{\pi \times \left\{ (d_{\text{Bump}}/2)^2 - (d_{\text{TSV}}/2 + t_{\text{ox},\text{bot}})^2 \right\}}{t_{\text{ox},\text{bot}}} [F].$$

➡ Bump02 Capacitance is a Parallel Capacitance

➡ Bump - Bottom - Silicon Substrate

➡ The General Formula for a Cylindrical Capacitance is

$$C = \frac{\varepsilon_0 A}{d}$$

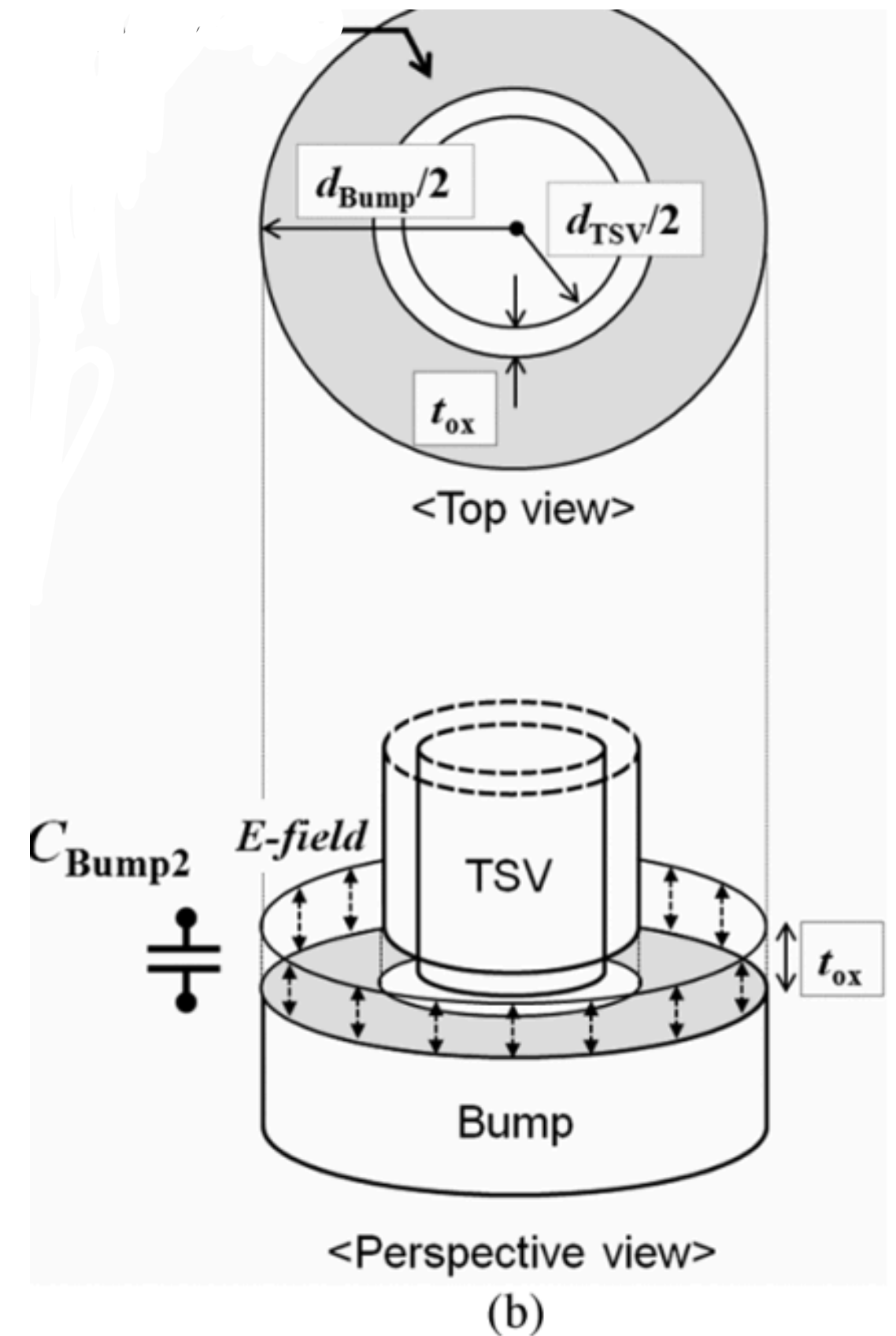


Fig.6.b. A top view and a perspective view of the bottom part of the TSV ⁽¹⁾

Underfill Capacitance

$$C_{\text{Underfill}} = \frac{\pi \times \epsilon_0 \epsilon_{r,\text{Underfill}}}{\cosh^{-1} \left(\frac{p_{\text{TSV}}}{d_{\text{Bump}}} \right)} \times h_{\text{Bump}} [F]$$

- Underfill Capacitance is the capacitance between two parallel capacitances
- Bump - Underfill - Bump
- The General Formula for a Parallel Cylinders Capacitance is

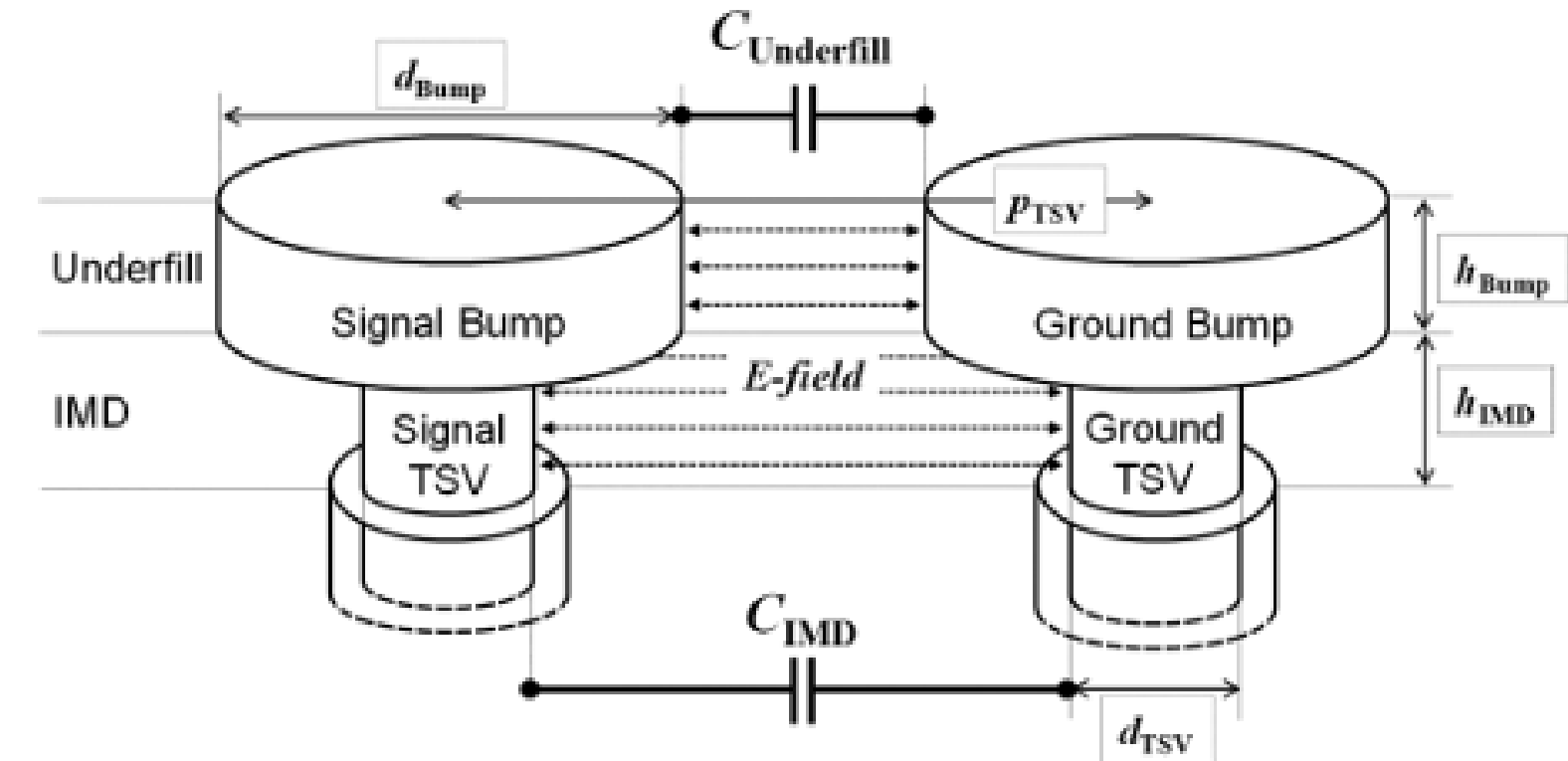


Fig.7.a. Underfill Capacitance between two bumps⁽¹⁾

$$C = \frac{\pi \epsilon L}{\cosh^{-1} \left(\frac{d}{2a} \right)}$$

Fig.7.b. Formula for a Parallel Plate capacitor

IMD ; Bottom ; Si Substrate Capacitance

Similar to the underfill capacitance IMD, Bottom, Si Substrate capacitances are the capacitance between two parallel cylinders.

IMD Capacitance: [TSV - IMD - TSV]

$$C_{\text{IMD}} = \frac{\pi \times \epsilon_0 \epsilon_{r,\text{IMD}}}{\cosh^{-1} \left(\frac{p_{\text{TSV}}}{d_{\text{TSV}}} \right)} \times h_{\text{IMD}} [F]$$

Bottom Capacitance: [TSV - Bottom - TSV]

$$C_{\text{Bottom}} = \frac{\pi \times \epsilon_0 \epsilon_{r,\text{ox,bot}}}{\cosh^{-1} \left(\frac{p_{\text{TSV}}}{d_{\text{TSV}}} \right)} \times t_{\text{ox,bot}} [F].$$

Si Sub Capacitance: [TSV - Si Substrate - TSV]

$$C_{\text{Si sub}} = \frac{\pi \times \epsilon_0 \epsilon_{r,\text{Si}}}{\cosh^{-1} \left(\frac{p_{\text{TSV}}}{d_{\text{TSV}}} \right)} \times (h_{\text{TSV}} - h_{\text{IMD}}) [F].$$

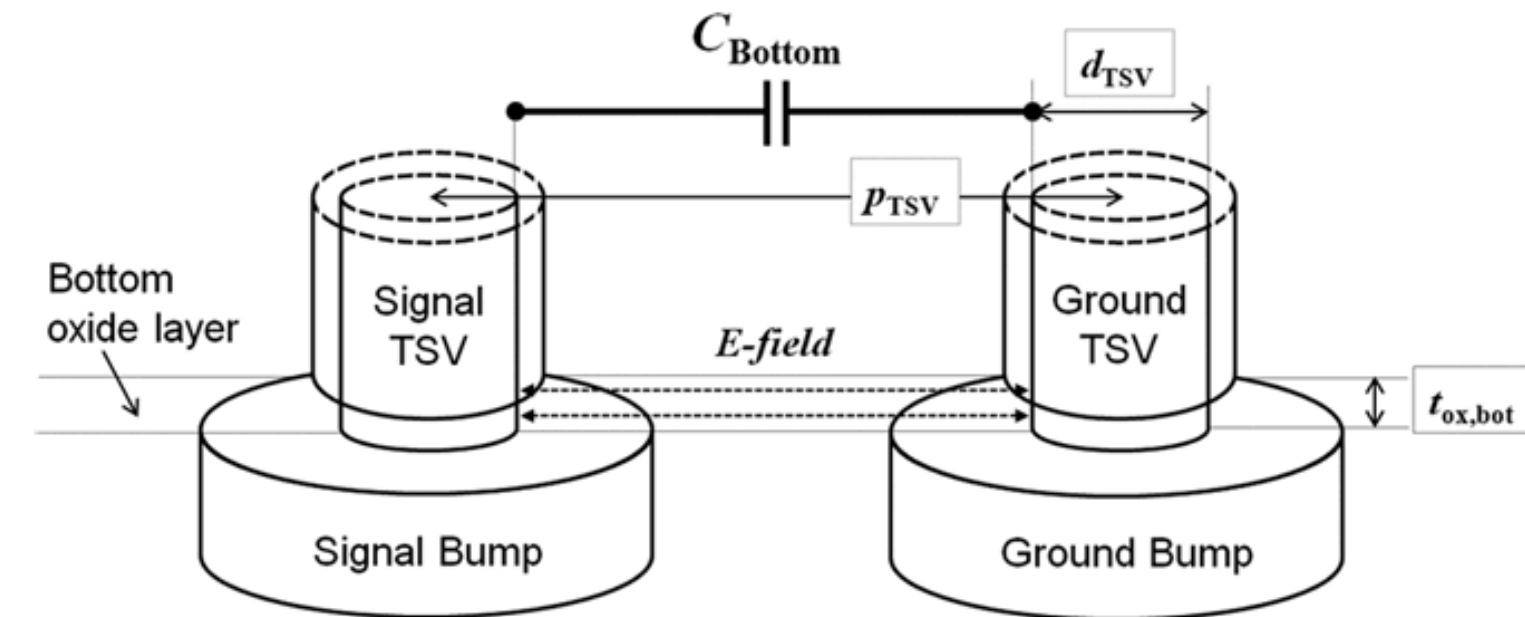


Fig.8.a. Bottom capacitance between GTSV and STSV⁽¹⁾

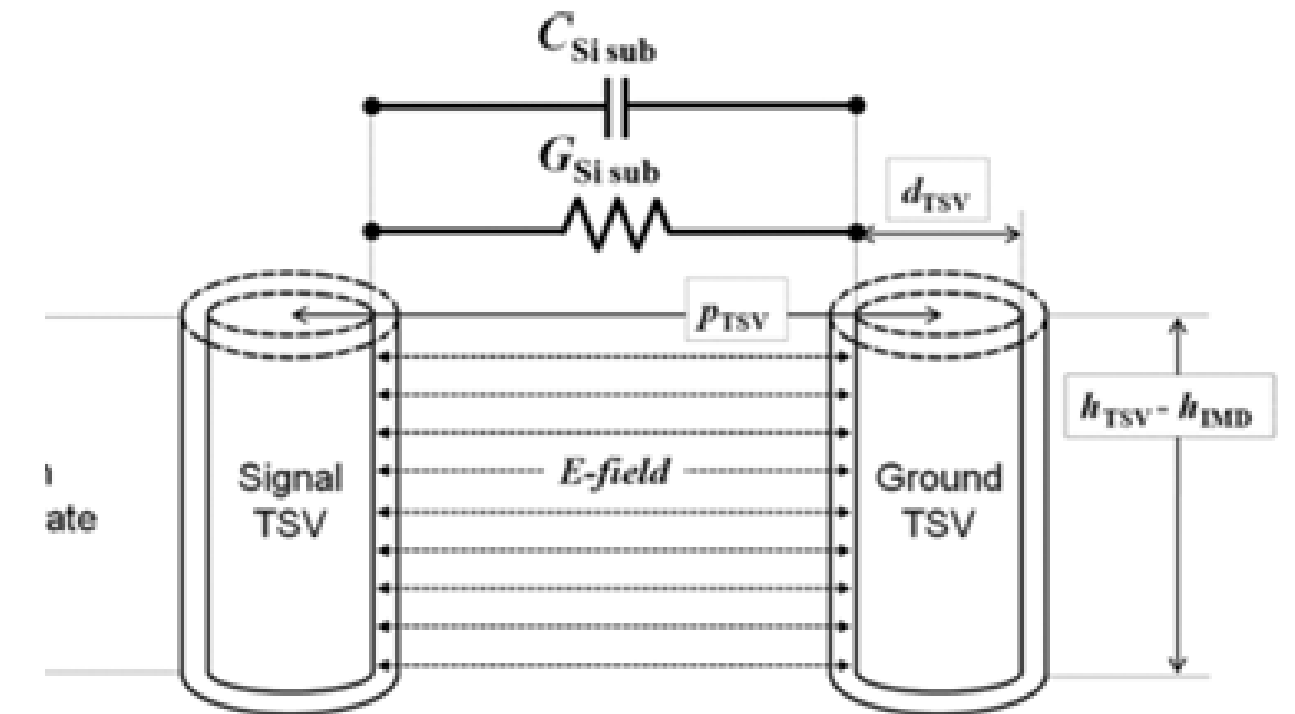


Fig.8.b. si substrate capacitance between GTSV and STSV⁽¹⁾

Inductance of TSV

Self Inductance:

- General formula for self inductance of a cylinder

$$L = \left(\frac{\mu_o \mu_r l}{2\pi} \right) \ln \left(\frac{b}{a} \right) [H]$$

- The Self Inductance of Single TSV:

$$L_{TSV} = \frac{1}{2} \left(\frac{\mu_o \mu_{r,TSV} h_{TSV}}{2\pi} \right) \ln \left(\frac{p_{TSV}}{r_{TSV}} \right) [H]$$

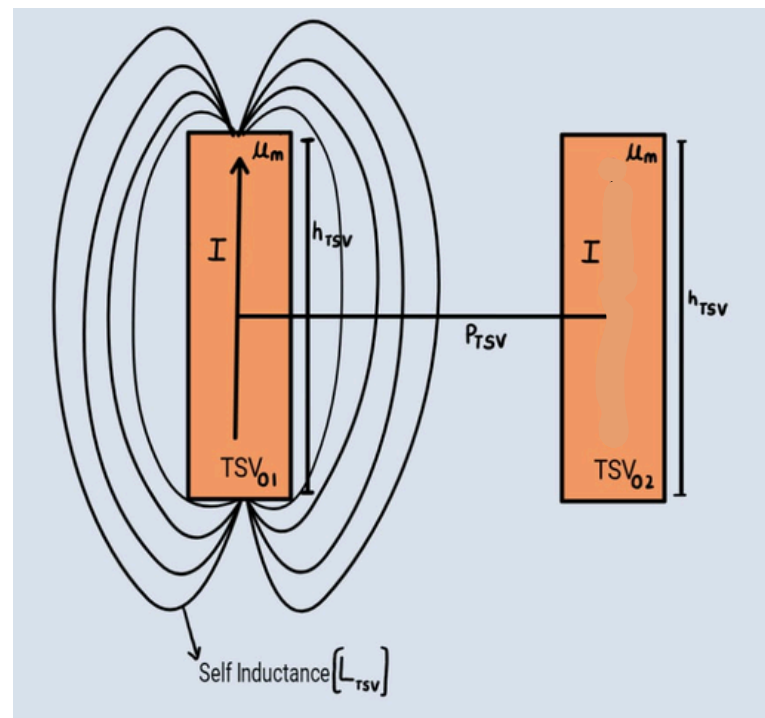


Fig.9.a Self Inductance of Single TSV

Mutual Inductance:

- General formula for mutual inductance between two cylinders: $L_M = \left(\frac{\mu_o \mu_r l}{2\pi} \right) \left(\ln \left(\frac{2l}{p} \right) - 1 \right) [H]$

- The Mutual Inductance of two TSV's:

$$L_M = \left(\frac{\mu_o \mu_{r,TSV}}{2\pi} \right) \left(\ln \left(\frac{2l_{TSV}}{p_{TSV}} \right) - 1 \right) [H]$$

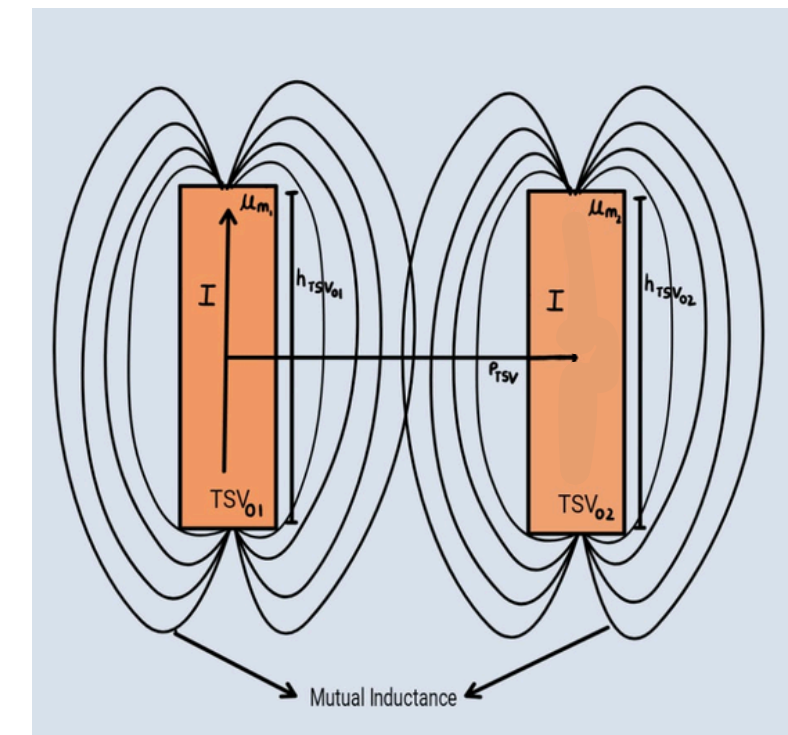


Fig.9.b. Mutual Inductance of two TSV

Resistance of TSV

$$R_{\text{TSV}} = \sqrt{(R_{\text{dc,TSV}})^2 + (R_{\text{ac,TSV}})^2} [\Omega]$$

Where -

$$R_{\text{dc,TSV}} = \rho_{\text{TSV}} \times \frac{h_{\text{TSV}}}{\pi \times \left(\frac{d_{\text{TSV}}}{2}\right)^2} [\Omega] \quad \& \quad R_{\text{ac,TSV}} = k_p \left(\rho_{\text{TSV}} \times \frac{h_{\text{TSV}}}{2\pi \times \frac{d_{\text{TSV}}}{2} \times \delta_{\text{skin depth,TSV}} - \pi \delta_{\text{skin depth,TSV}}^2} \right) [\Omega]$$

$$\text{Where - } \delta_{\text{skindepth,TSV}} = \frac{1}{\sqrt{\pi f \mu_{\text{TSV}} \sigma_{\text{TSV}}}} [\text{m}]$$

Conductance of TSV

$$\text{W.K.T } C = \varepsilon K \quad \& \quad G = \sigma K \quad \text{Where - K is the geometry factor}$$

$$\frac{C_{\text{Si sub}}}{G_{\text{Si sub}}} = \frac{\varepsilon_{\text{Si}}}{\sigma_{\text{Si}}}$$
$$G_{\text{Si sub}} = \frac{\pi \times \sigma_{\text{Si}}}{\cosh^{-1} \left(\frac{p_{\text{TSV}}}{d_{\text{TSV}}} \right)} \times (h_{\text{TSV}} - h_{\text{IMD}}) [S].$$



Defected TSV Electrical Modelling

1. Pinhole Defect is Modelled in terms of RL
2. Pinhole Defect is Modelled in terms of C_{Leak} and G_{Leak}

Pinhole Defect in Terms of RL

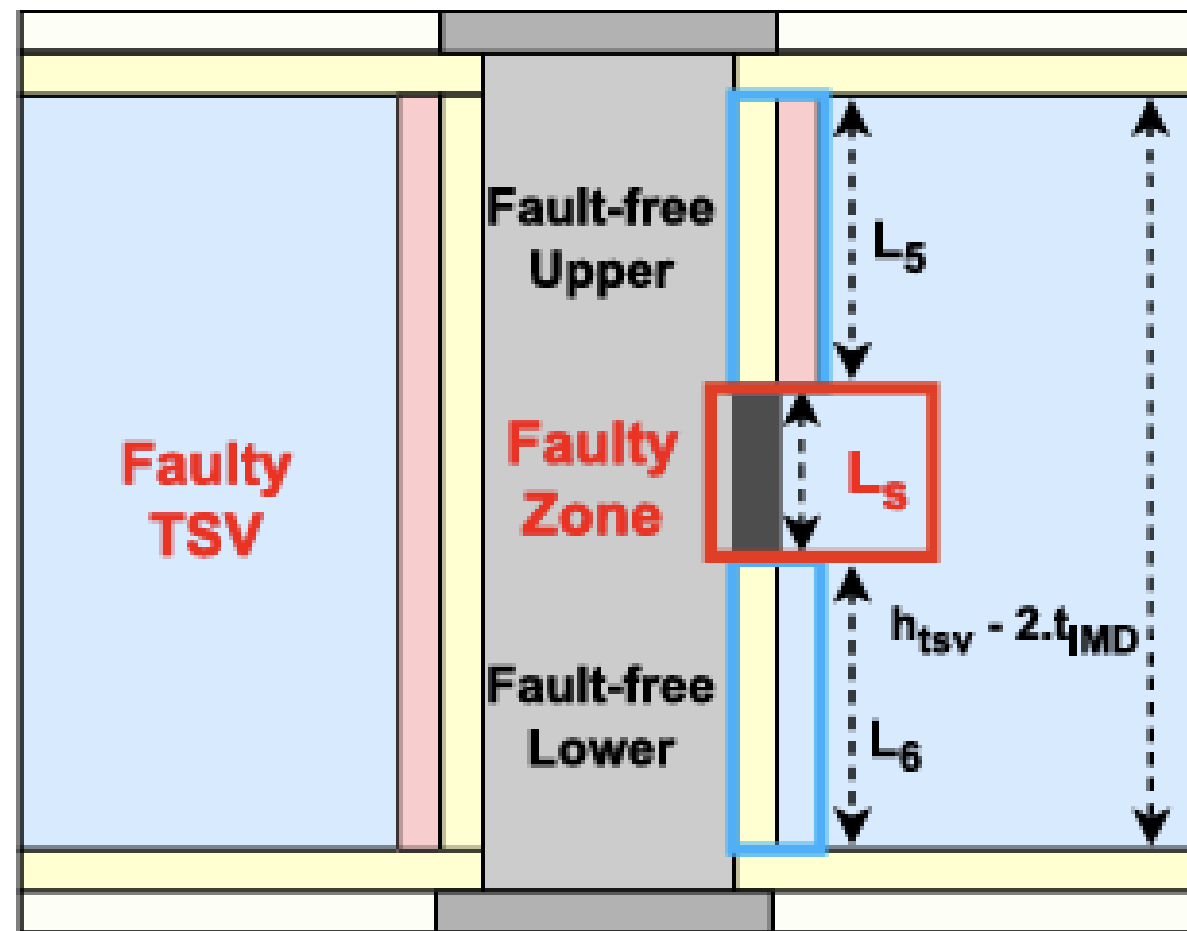


Fig.10.a. Schematic of TSV with short fault.⁽³⁾

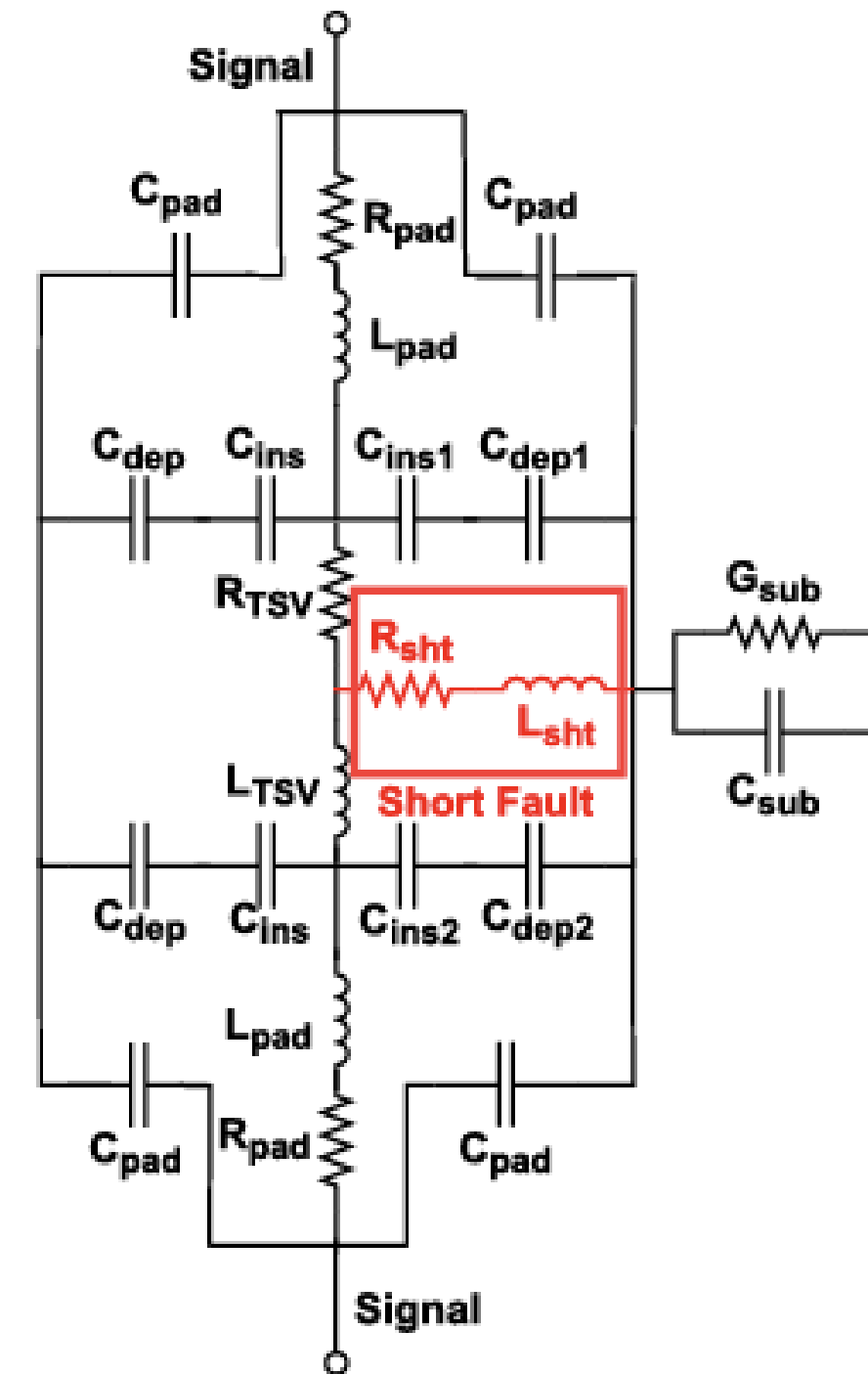


Fig.10.b. Equivalent circuit of TSV with short fault.⁽³⁾

📖 R,L Equations

➡
$$R_{\text{sht}} = \frac{\rho_T \cdot t_{\text{ins}}}{L_s^2}$$

➡
$$L_{\text{sht}} = \frac{\mu \cdot \mu_{\text{tsv}} \cdot t_{\text{ins}}}{2\pi} \cdot \left(\operatorname{arccosh} \left(\frac{t_{\text{ins}}}{\sqrt{L_s^2 + t_{\text{ins}}^2}} \right) - \sqrt{2 \cdot t_{\text{ins}}^2 + L_s^2} + \frac{t_{\text{ins}}}{4} + \sqrt{L_s^2 + t_{\text{ins}}^2} \right)$$

Pinhole Defect in Terms of C_{Leak} and G_{Leak}

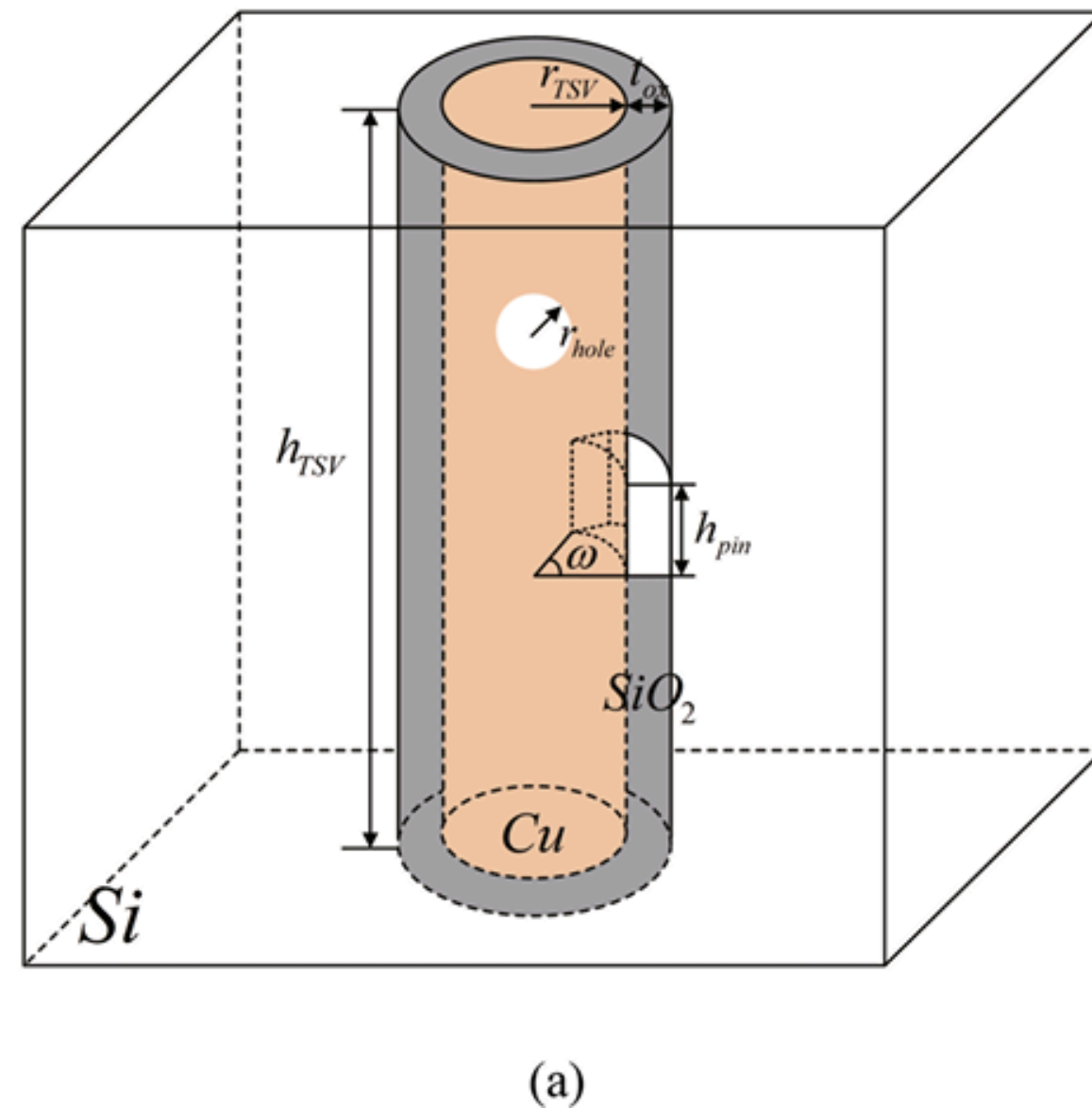


Fig.11.a. Defect TSV structure diagram⁽⁴⁾

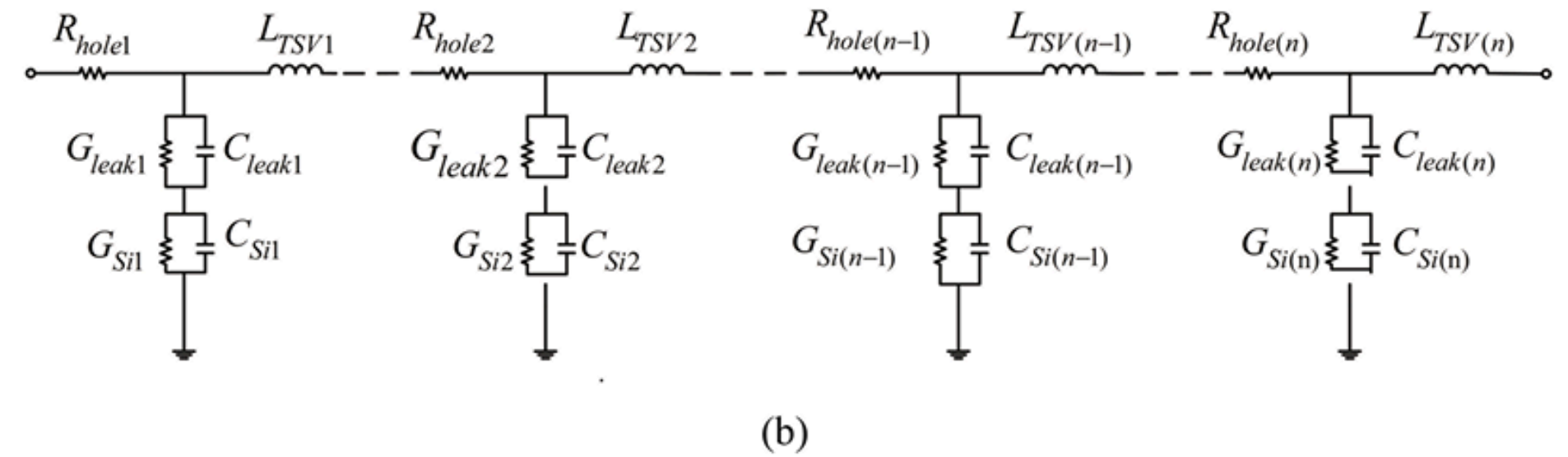


Fig.11.b. defect TSV equivalent circuit diagram⁽⁴⁾

Equations of C_{Leak} and G_{Leak}

➡
$$C_{leak} = \frac{2\pi (\epsilon_0 \epsilon_{r,ox} l_{TSV} - \epsilon_0 \epsilon_{r,air} j r_{TSV})}{\ln \left(\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right)} [F] \quad \text{Where} \quad j = \frac{\omega h_{pin}}{2\pi r_{TSV}}$$

$$C_{leak} = C_{ins} - C_{air}$$

$$\left[j = \frac{DamagedArea}{SideWallAreaAffected} \right]$$

➡
$$G_{leak} = \frac{2\pi \sigma_{SiO_2} l_{TSV} - 4\pi^2 j f \epsilon_0 \epsilon_{r,air} r_{TSV}}{\ln \left(\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right)}$$

$$G_{leak} = G_{ox} - Y_{air}$$

G_{air} is almost zero so we take air Admittive capacitance



References

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- ➡ {2} Chandrakar, M., Solanki, K., Majumder, M. K., & Sharma, R. Reliability Concerns in Through Silicon Via based 3D Integration: Fabrication to Packaging.
- ➡ {3} Kirtonia, P., Williams, S., Akter, S., Bayoumi, M., & Khalil, K. (2025). A Novel TSV Model With Fault Characterization for High-Frequency Transmission in 3D ICs. IEEE Transactions on Circuits and Systems I: Regular Papers.
- ➡ {4} Liu, C., Dong, G., Zhi, C., & Zhu, Z. (2025). Modelling and simulation of TSV considering void and leakage defects. Journal of Computational Electronics, 24(2), 61.
- ➡ {5} Shang, Y., Tan, M., Li, C., & Sun, L. (2019). TSV manufacturing fault modelling and diagnosis based on multi-tone dither. Journal of Advanced Computational Intelligence and Intelligent Informatics, 23(1), 42-51.